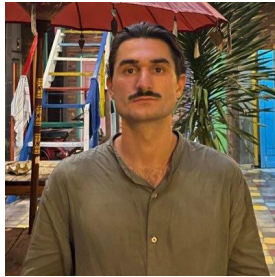


Learning Predictive Embeddings through Inter-View Regressor Alignment

Learning Predictive Embeddings through Inter-View Regressor Alignment



Basile Terver

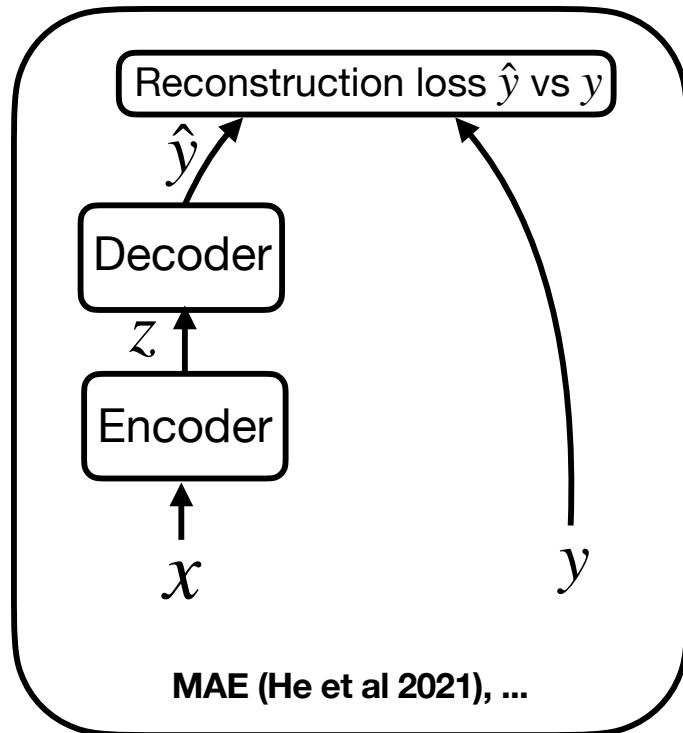


Jean Ponce

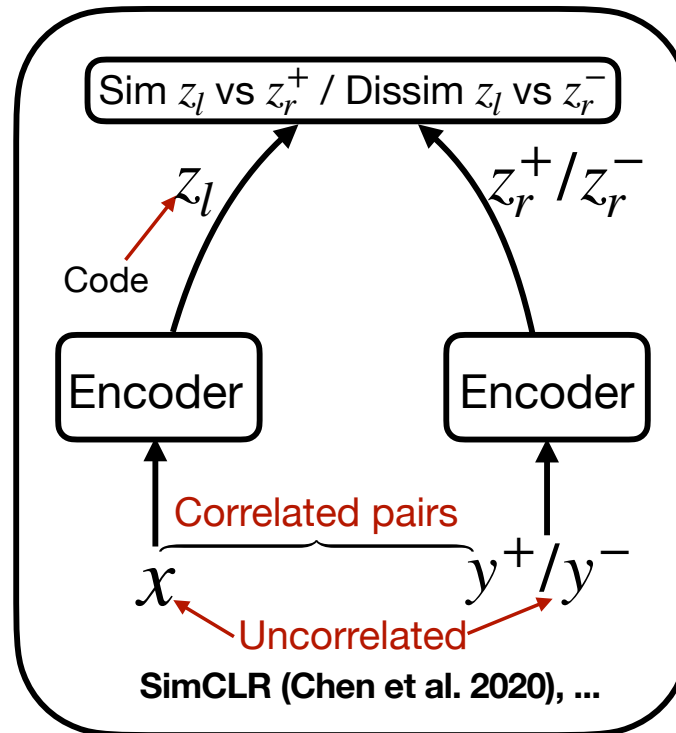
Self-supervised learning

Learn data representation without supervision by comparing altered/partial views of the same input

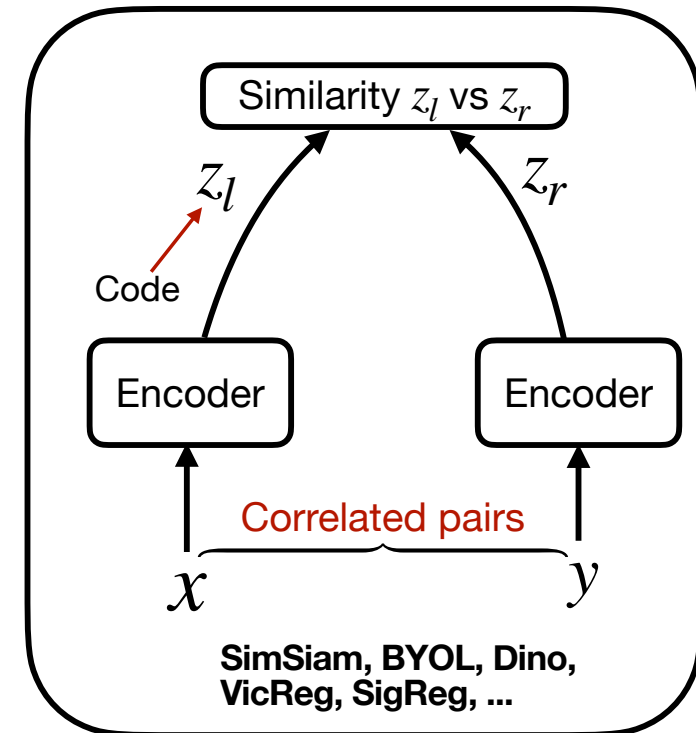
Reconstruction-based Methods



Contrastive SSL



Non-Contrastive SSL



SimSiam-like dynamics for SSL

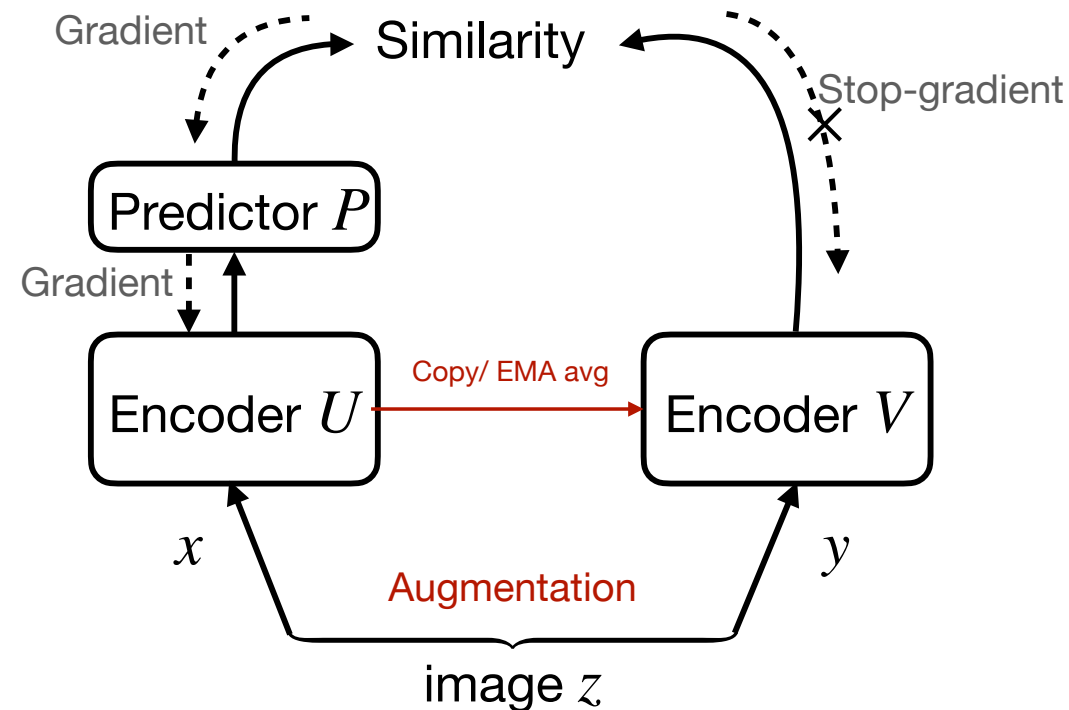
What is known

SimSiam-like SSL methods: A puzzle

- Strong empirical performance
- Heuristic asymmetry (Stop-gradient, EMA)
- Avoid collapse in some cases

Goal: Understanding:

- What is learned?
- When does collapse occur?
- How to reliably prevent collapse?



SimSiam-like SSL: Existing theory is informative but restricted

Special setting

- Linear encoders U, V
- Particular data covariance structure
- Special initialization (ex: orthogonal)

Tian et al. 2021

Littwin et al. 2024

Liu et al. 2022

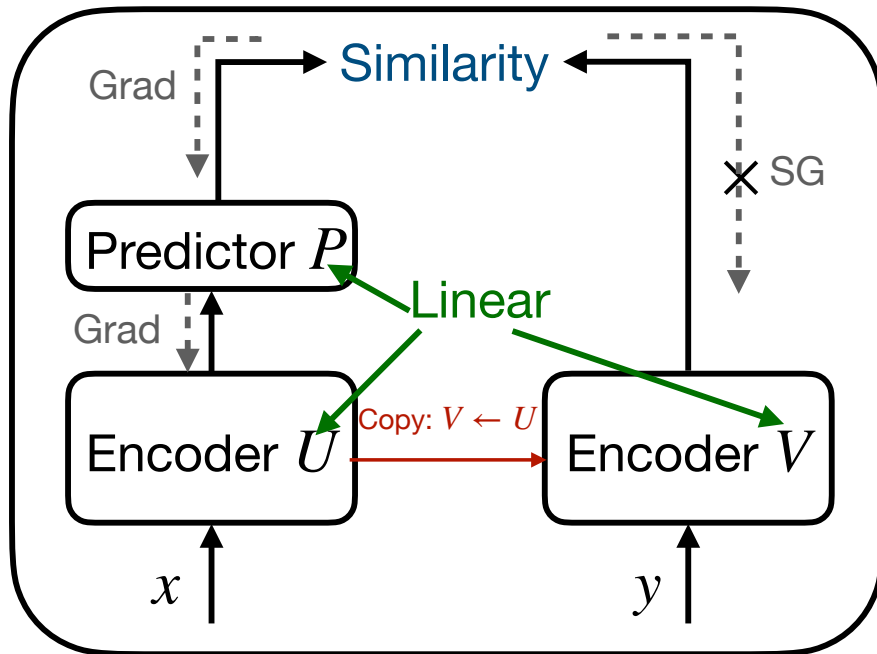
Tang et al. 2023

?

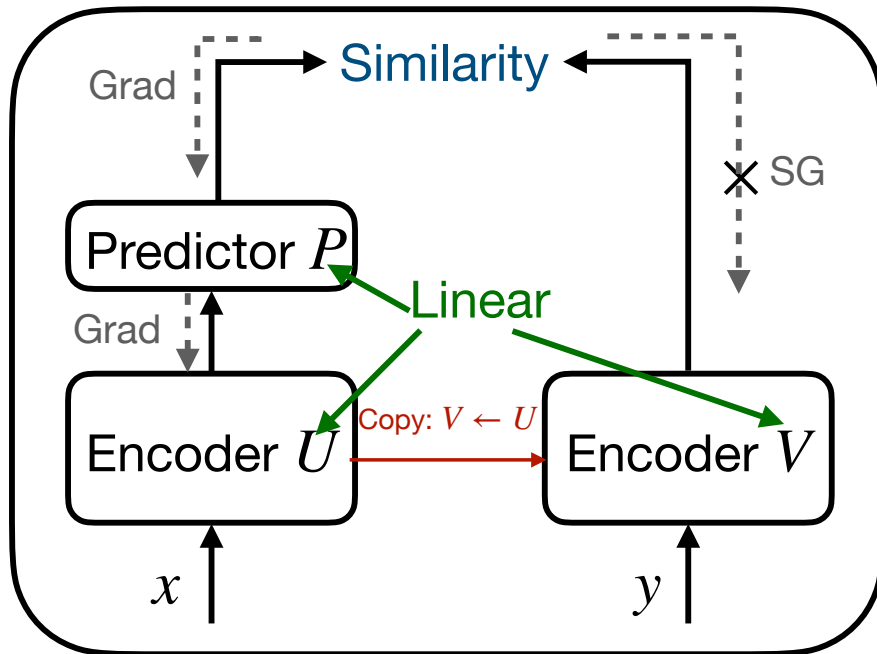
General setting

- Non-linear encoders U, V
- General data distributions
- General initialization

SimSiam-like SSL: Linear encoder and predictor



SimSiam-like SSL: Linear encoder and predictor



Setting considered in Tian et al. 2021, etc

Similarity

$$\mathcal{L}(P, U, V) = \frac{1}{2} \mathbb{E}[\|PUx - Vy\|^2] + \frac{\lambda}{2} \|P\|^2 + \frac{\lambda}{2} \|U\|^2$$

Continuous-time dynamics

$$\begin{cases} \dot{P} = -\partial_P \mathcal{L}(P, U, V) \\ \dot{U} = -\partial_U \mathcal{L}(P, U, V) \\ V = U \end{cases}$$

Not the gradient flow of any objective
(Ponce et al. 2025)

Linear encoder and optimal predictor (Tang et al. 2023)

$$\mathcal{L}(P, U, V) = \frac{1}{2} \mathbb{E}[\|PUx - Vy\|^2] + \frac{\lambda}{2} \|P\|^2 + \frac{\lambda}{2} \|U\|^2$$

$$\begin{cases} \dot{P} = -\partial_P \mathcal{L}(P, U, V) \\ \dot{U} = -\partial_U \mathcal{L}(P, U, V) \\ V = U \end{cases}$$

Optimal predictor

$$\begin{cases} P^* = \arg \min_P \mathcal{L}(P, U, V) \\ \dot{U} = -\partial_U \mathcal{L}(P^*, U, V) \\ V = U \end{cases}$$

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Theorem (Adapted)

Assume:

- Orthogonal initialization $U_0 U_0^\top = I \in \mathbb{R}^{k \times k}$
- Matrix $\Sigma_{xx} = \mathbb{E}[xx^\top] = \alpha I$
- Matrix $\Sigma_{yx} = \mathbb{E}[yx^\top]$ is symmetric

Then:

- U stays orthogonal over time
- $\mathcal{F}(U) = -\frac{1}{2} \text{Tr}((U \Sigma_{yx} U^\top)^2)$ **decreases** over time
- Rows of an equilibrium U^* span
an eigen-subspace of Σ_{yx} of at most k dimensions

Linear encoder and optimal predictor (Tang et al. 2023)

$$\mathcal{L}(P, U, V) = \frac{1}{2} \mathbb{E}[\|PUx - Vy\|^2] + \frac{\lambda}{2} \|P\|^2 + \frac{\lambda}{2} \|U\|^2$$

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Q: Convergence to an equilibrium U^* ? Which eigen-subspace of Σ_{yx} is selected?

A: Unknown

Linear encoder and optimal predictor (Tang et al. 2023)

$$\mathcal{L}(P, U, V) = \frac{1}{2} \mathbb{E}[\|PUx - Vy\|^2] + \frac{\lambda}{2} \|P\|^2 + \frac{\lambda}{2} \|U\|^2$$

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Q: General initialization? General data? (Still with linear encoders)

A: Complex dynamical system, hard to analyze (Ponce et al. 2025)

Linear encoder and optimal predictor (Tang et al. 2023)

$$\mathcal{L}(P, U, V) = \frac{1}{2} \mathbb{E}[\|PUx - Vy\|^2] + \frac{\lambda}{2} \|P\|^2 + \frac{\lambda}{2} \|U\|^2$$

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- Rows of an equilibrium U^* span
an eigen-subspace of Σ_{yx} of at most k dimensions

Q: What about non-linear encoders?

A: Seems even harder, but not so if done in function spaces

**Non-linear SimSiam-like dynamics:
New convergence results**

A SimSiam-like dynamics in function space

$$\mathcal{L}(P, U, V) = \frac{1}{2}\mathbb{E}[\|PU(x) - V(y)\|^2] + \frac{\lambda}{2}\|P\|^2 + \frac{\lambda}{2}\mathbb{E}[\|U(x)\|^2]$$

$$\begin{cases} P^\star = \arg \min_P \mathcal{L}(P, U, V) \\ \dot{U} = -\partial_U \mathcal{L}(P^\star, U, V) \\ V = U \end{cases}$$

- Optimal linear predictor (as in Tang et al. 2023)
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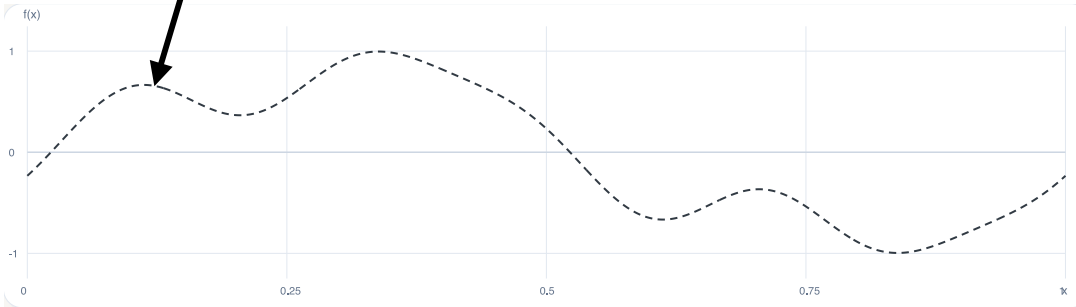
Why care about function spaces?

Aren't we training a neural network after all?

Why care about function spaces?

Neural Networks are function approximators

Target $f^\star$

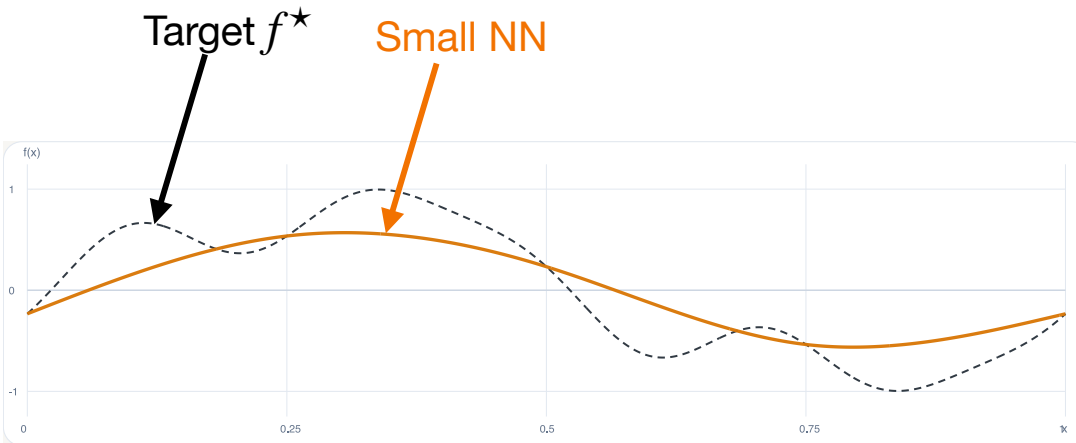


Regression: $f^\star = \arg \min_f \mathbb{E} [\|y - f(x)\|^2]$

Target: $f^\star(x) = \mathbb{E} [y | x]$

Why care about function spaces?

Neural Networks are function approximators

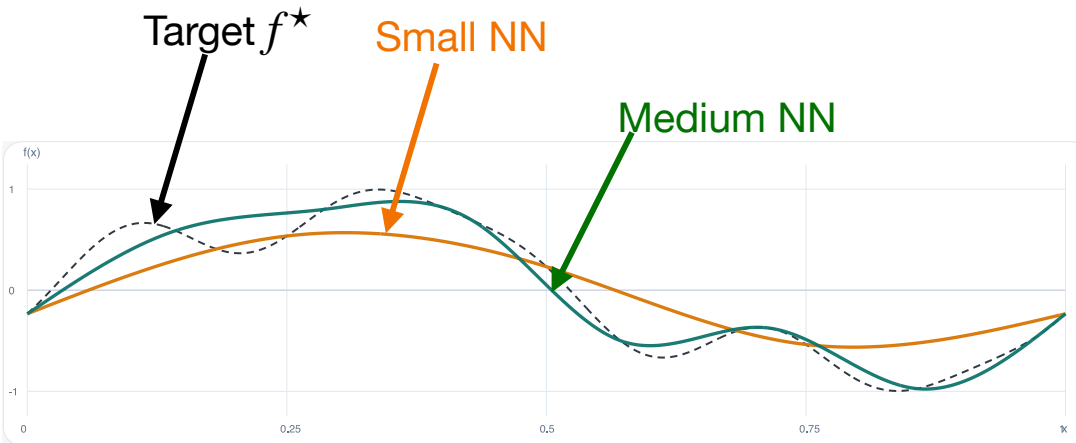


Regression:
$$f^* = \arg \min_f \mathbb{E} [\|y - f(x)\|^2]$$

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Why care about function spaces?

Neural Networks are function approximators

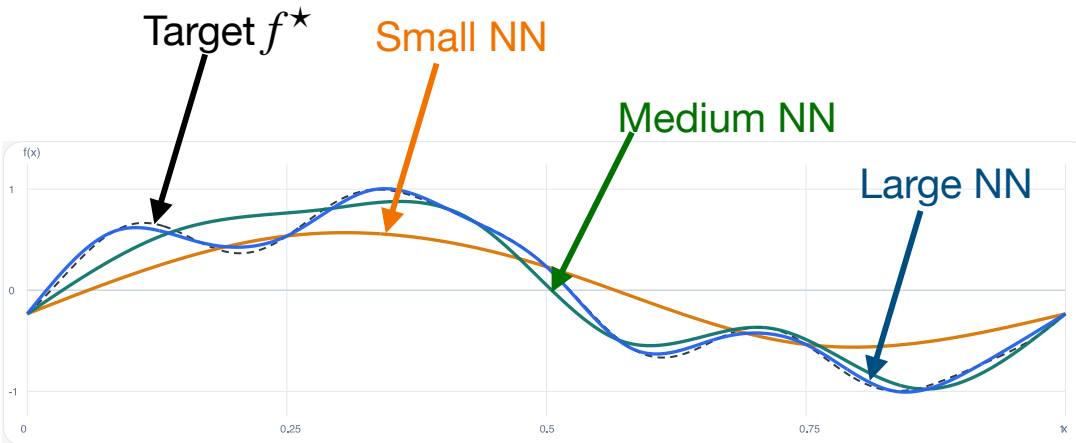


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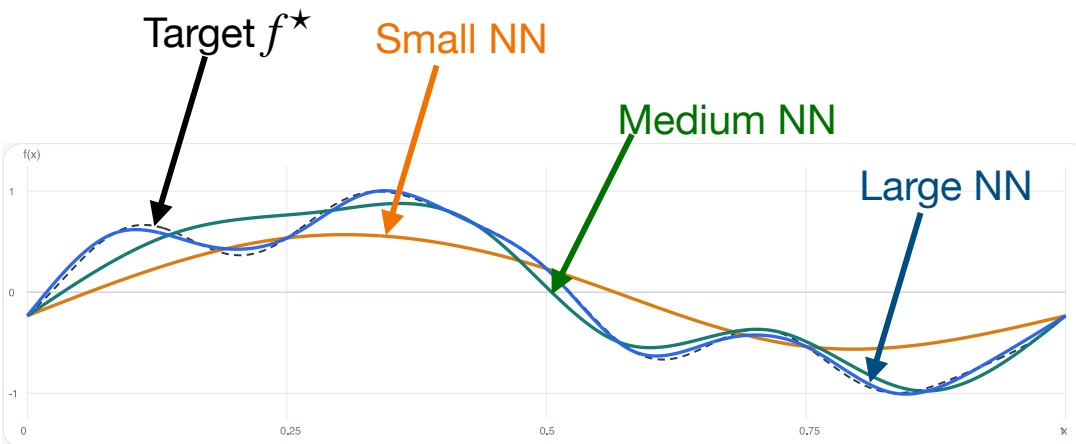


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$$f^* = \arg \min_f \mathbb{E} [\|y - f(x)\|^2]$$

Target:
$$f^*(x) = \mathbb{E} [y | x]$$

Why care about function spaces?

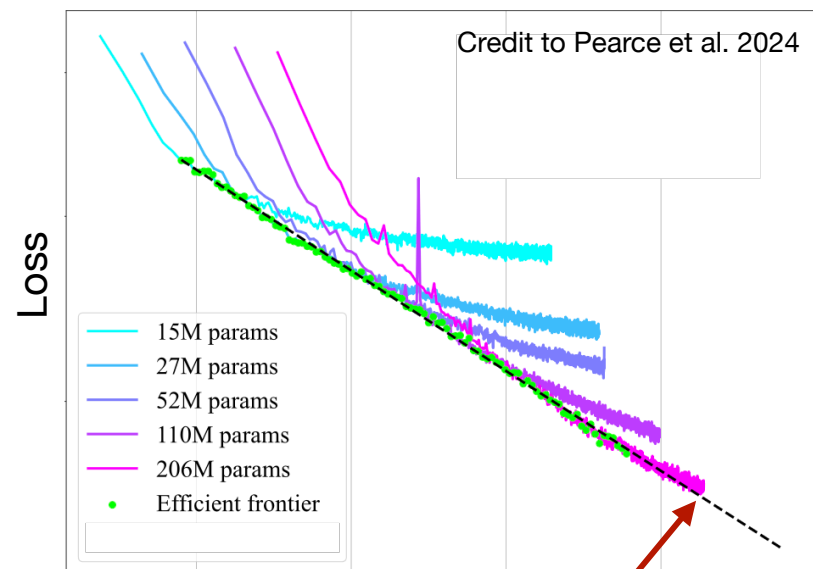
Neural Networks are function approximators



Regression: $f^* = \arg \min_f \mathbb{E} [\|y - f(x)\|^2]$

Target: $f^*(x) = \mathbb{E} [y | x]$

Scale expressivity, data and compute



FLOPs

Approach a functional target

Function space analysis reveals the target (properties, stability, etc)

A SimSiam-like dynamics in function space

$$\mathcal{L}(P, U, V) = \frac{1}{2}\mathbb{E}[\|PU(x) - V(y)\|^2] + \frac{\lambda}{2}\|P\|^2 + \frac{\lambda}{2}\mathbb{E}[\|U(x)\|^2]$$

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- No parametric form for the encoder U
- Dynamics defined in function space $L_2(\mathbb{P}_x, \mathbb{R}^k)$

Why care about function spaces?

Function space analysis reveals the target (properties, stability, etc)

A SimSiam-like dynamics in function space: Convergence

$$\mathcal{L}(P, U, V) = \frac{1}{2}\mathbb{E}[\|PU(x) - V(y)\|^2] + \frac{\lambda}{2}\|P\|^2 + \frac{\lambda}{2}\mathbb{E}[\|U(x)\|^2]$$

$$\begin{cases} P^\star = \arg \min_P \mathcal{L}(P, U, V) \\ \dot{U} = -\partial_U \mathcal{L}(P^\star, U, V) \\ V = U \end{cases}$$

- Optimal linear predictor (as in Tang et al. 2023)
- No parametric form for the encoder U
- Dynamics defined in function space $L_2(\mathbb{P}_x, \mathbb{R}^k)$

Theorem (Informal): Assume x and y play symmetric roles and data regular enough.

Then: 1- For any initialization U_0 , the dynamics converges to an equilibrium $U^\star$.

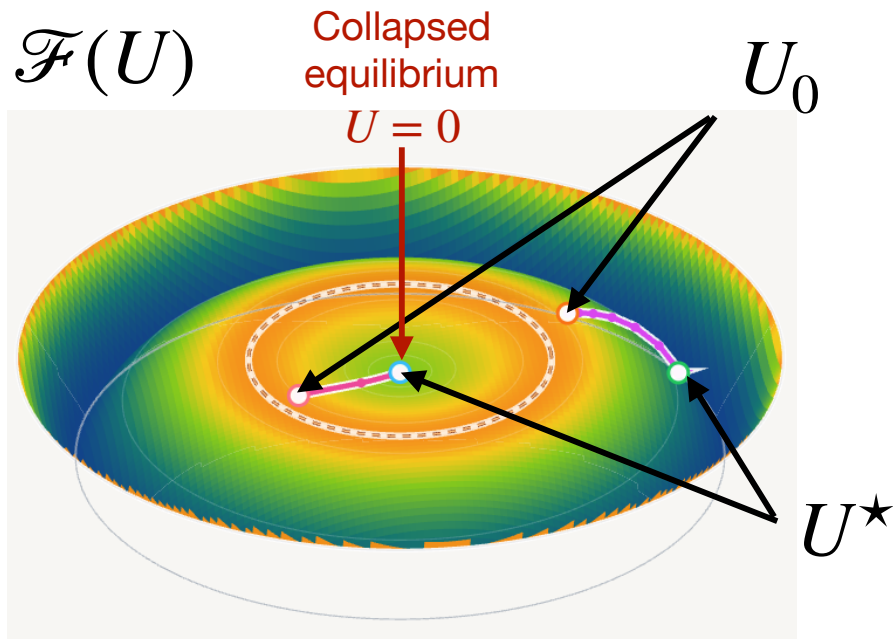
2- The following objective decreases in time (Lyapunov function):

Recovers the objective
from Tang et al. 2023
under their assumptions

$$\mathcal{F}(U) = -\frac{1}{2}\text{Tr}(\mathbb{E}[U(x)U(y)^\top]^2) + \frac{1}{3}\text{Tr}((\mathbb{E}[U(x)U(x)^\top] + \lambda I)^3)$$

No assumption on the data covariances, or orthogonal initialization, as in Tang et al. 2023

A SimSiam-like dynamics in function space: Convergence



- No limit cycles, no divergence.
- Convergence to a critical point of the Lyapunov function.

Key technical ingredients

- **Gradient-like system:** The vector field aligns well with the gradient of the Lyapunov function
- **Lojasiewicz-Simon inequality:** Small gradients $\implies$ Small gap to a critical value.
- **Trapping argument:** The trajectory is trapped near an accumulation point (Bolte et al. 2014)
- **Existence of accumulation points:** Non-trivial in infinite dimensions (Palais & Smale 1964)

A SimSiam-like dynamics in function space: Equilibria

$$\mathcal{L}(P, U, V) = \frac{1}{2}\mathbb{E}[\|PU(x) - V(y)\|^2] + \frac{\lambda}{2}\|P\|^2 + \frac{\lambda}{2}\mathbb{E}[\|U(x)\|^2]$$

$$\begin{cases} P^\star = \arg \min_P \mathcal{L}(P, U, V) \\ \dot{U} = -\partial_U \mathcal{L}(P^\star, U, V) \\ V = U \end{cases}$$

- Optimal linear predictor (as in Tang et al. 2023)
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- Dynamics defined in function space $L_2(\mathbb{P}_x, \mathbb{R}^k)$

What are the equilibria? Which ones are stable?

CCA to the rescue

Canonical Correlation Analysis

Linear Canonical Correlation Analysis

Correlation maximization

$$\max_{U, V \in \mathbb{R}^{k \times d}} \mathbb{E}[(Ux)^{\top}(Vy)]$$

s.t. $\mathbb{E}[(Ux)(Ux)^{\top}] = I$

$$\mathbb{E}[(Vy)(Vy)^{\top}] = I$$

whitening constraints

Linear Canonical Correlation Analysis

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$$\mathbb{E}[(Vy)(Vy)^{\top}] = I$$

whitening constraints

Whitened cross-covariance

$$M = \Sigma_{yy}^{-\frac{1}{2}} \Sigma_{yx} \Sigma_{xx}^{-\frac{1}{2}}$$

$\mathbb{E}[yy^{\top}] \quad \mathbb{E}[yx^{\top}] \quad \mathbb{E}[xx^{\top}]$

Singular value decomposition

$$M = (v_1, \dots, v_d) \begin{pmatrix} s_1 & & \\ & \ddots & \\ & & s_d \end{pmatrix} \begin{pmatrix} u_1^{\top} \\ \vdots \\ u_d^{\top} \end{pmatrix}$$

CCA solution: top-k singular vector

$$U = \Sigma_{xx}^{-\frac{1}{2}} \begin{pmatrix} u_1 \\ \vdots \\ u_k \end{pmatrix}, \quad V = \Sigma_{yy}^{-\frac{1}{2}} \begin{pmatrix} v_1 \\ \vdots \\ v_k \end{pmatrix}$$

Linear Canonical Correlation Analysis

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Recovers similar eigen-subspace as Tang et al. 2023
under their assumptions

(Σ_{yx} symmetric and $\Sigma_{xx} = \Sigma_{yy} = \alpha I$)

Non-Linear Canonical Correlation Analysis (Michaeli et al. 2015)

Correlation maximization

$$\max_{(U,V) \in \mathcal{U} \times \mathcal{V}} \mathbb{E}[U(x)^\top V(y)]$$

$L_2(\mathbb{P}_x)^k$

$L_2(\mathbb{P}_y)^k$

s.t.

$$\mathbb{E}[U(x)U(x)^\top] = I$$

$$\mathbb{E}[V(y)V(y)^\top] = I$$

whitening constraints

Non-Linear Canonical Correlation Analysis (Michaeli et al. 2015)

Correlation maximization

$$\max_{(U,V) \in \mathcal{U} \times \mathcal{V}} \mathbb{E}[U(x)^\top V(y)]$$

$L_2(\mathbb{P}_x)^k$ $L_2(\mathbb{P}_y)^k$

s.t. $\mathbb{E}[U(x)U(x)^\top] = I$
 $\mathbb{E}[V(y)V(y)^\top] = I$

whitening constraints

Non-linear CCA solution: top-k canonical functions

$$U(x) = \begin{pmatrix} u_1(x) \\ \vdots \\ u_k(x) \end{pmatrix}, \quad V(y) = \begin{pmatrix} v_1(y) \\ \vdots \\ v_k(y) \end{pmatrix}$$

Canonical correlation 1 = $s_1 \geq s_2 \geq \dots$

$$\mathbb{E}[u_i(x)v_i(y)] = s_i \quad \mathbb{E}[u_i(x)v_j(y)] = 0$$

De-correlation (when $i \neq j$)

Orthonormal basis of $L_2(\mathbb{P}_y)$

Orthonormal basis of $L_2(\mathbb{P}_x)$

Non-Linear Canonical Correlation Analysis (Michaeli et al. 2015)

Correlation maximization

$$\max_{(U,V) \in \mathcal{U} \times \mathcal{V}} \mathbb{E}[U(x)^\top V(y)]$$

$L_2(\mathbb{P}_x)^k$ $L_2(\mathbb{P}_y)^k$

s.t. $\mathbb{E}[U(x)U(x)^\top] = I$
 $\mathbb{E}[V(y)V(y)^\top] = I$

whitening constraints

Connection with VICReg, BarlowTwins
(Balestriero & LeCun 2022)

What about SimSiam with nonlinear encoder ?

Non-linear CCA solution: top-k canonical functions

$$U(x) = \begin{pmatrix} u_1(x) \\ \vdots \\ u_k(x) \end{pmatrix}, \quad V(y) = \begin{pmatrix} v_1(y) \\ \vdots \\ v_k(y) \end{pmatrix}$$

Canonical correlation $1 = s_1 \geq s_2 \geq \dots$

$$\mathbb{E}[u_i(x)v_i(y)] = s_i \quad \mathbb{E}[u_i(x)v_j(y)] = 0$$

De-correlation (when $i \neq j$)
Orthonormal basis of $L_2(\mathbb{P}_y)$
Orthonormal basis of $L_2(\mathbb{P}_x)$

Non-linear SimSiam-like dynamics:

Equilibria and their stability

A SimSiam-like dynamics in function space: Equilibria

$$\mathcal{L}(P, U, V) = \frac{1}{2}\mathbb{E}[\|PU(x) - V(y)\|^2] + \frac{\lambda}{2}\|P\|^2 + \frac{\lambda}{2}\mathbb{E}[\|U(x)\|^2]$$

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- Optimal linear predictor (as in Tang et al. 2023)
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What are the equilibria? Which ones are stable?

Equilibria span CCA subspaces

SimSiam-like dynamics in function space: Stability of equilibria

Theorem (Informal):

Equilibria span CCA subspaces. The only stable ones are of the form:

$$U^\star(x) = \sum_{i=1}^r f(s_i, \lambda) q_i u_i(x)$$

Any $0 \leq r \leq k$

Regularization

Any orthonormal family $(q_i)_{1 \leq i \leq r}$ of $\mathbb{R}^k$

Canonical function

Spectral filter

Canonical correlation $s_1 \geq s_2 \geq \dots$

$$f(s, \lambda) = \frac{1}{\sqrt{2}}(s + \sqrt{s^2 - 4\lambda}) \mathbf{1}_{s^2 \geq 4\lambda}$$

SimSiam-like dynamics in function space: Stability of equilibria

Theorem (Informal):

Equilibria span CCA subspaces. The only stable ones are of the form:

$$U^\star(x) = \sum_{i=1}^r f(s_i, \lambda) q_i u_i(x)$$

Diagram illustrating the components of the equilibrium form:

- Any $0 \leq r \leq k$** : Points to the summation index r .
- Regularization**: Points to the parameter λ .
- Any orthonormal family $(q_i)_{1 \leq i \leq r}$ of $\mathbb{R}^k$** : Points to the vector q_i .
- Canonical function**: Points to the function $u_i(x)$.
- Canonical correlation $s_1 \geq s_2 \geq \dots$** : Points to the parameter s_i .
- Spectral filter**: Points to the function $f(s, \lambda)$.

The spectral filter function is defined as:

$$f(s, \lambda) = \frac{1}{\sqrt{2}}(s + \sqrt{s^2 - 4\lambda}) \mathbf{1}_{s^2 \geq 4\lambda}$$

Collapse occurs when an admissible CCA direction ($f(s_i, \lambda) > 0$ for $i \leq k$) is not selected ($r < i$)

Predictive Embeddings through Inter-View Regressor Alignment

Intuition and Motivation: Trace maximization of a predictor

- Optimal predictor between two views:

The diagram shows the formula for the optimal predictor $P^\star$ with three annotations:

- Shared signal:** A red arrow points to the term $\mathbb{E} [V(y)U(x)^\top]$, which is highlighted in a purple box.
- Within view variability (noise):** A blue arrow points to the term $\mathbb{E} [U(x)U(x)^\top]$ inside the denominator, which is highlighted in a blue box.
- Signal to noise ratio (matrix form):** A green arrow points to the entire denominator $\left(\mathbb{E} [U(x)U(x)^\top] + \lambda I \right)^{-1}$, which is highlighted in a green box.

$$P^\star = \mathbb{E} [V(y)U(x)^\top] \left(\mathbb{E} [U(x)U(x)^\top] + \lambda I \right)^{-1}$$

Intuition and Motivation: Trace maximization of a predictor

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$$P^\star = \mathbb{E} [V(y)U(x)^\top] \left(\mathbb{E} [U(x)U(x)^\top] + \lambda I \right)^{-1}$$

$Tr(P^\star)$ acts as a scalar summary of the matrix signal to noise ratio

Intuition and Motivation: Trace maximization of a predictor

- Stable equilibria of the SimSiam-like dynamics in function space

$$U^*(x) = \sum_{i=1}^r f(s_i, \lambda) q_i u_i(x)$$

Any $0 \leq r \leq k$

Regularization

Spectral filter

Canonical correlation $s_1 \geq s_2 \geq \dots$

The diagram illustrates the trace maximization formula for the optimal predictor $U^*(x)$. The formula is $U^*(x) = \sum_{i=1}^r f(s_i, \lambda) q_i u_i(x)$. Annotations include: a pink arrow pointing to the upper limit r with the text 'Any $0 \leq r \leq k$ '; a grey arrow pointing to λ with the text 'Regularization'; a blue arrow pointing to the function f with the text 'Spectral filter'; and a red arrow pointing to s_i with the text 'Canonical correlation $s_1 \geq s_2 \geq \dots$ '. The terms q_i and $u_i(x)$ are also present in the formula.

Intuition and Motivation: Trace maximization of a predictor

- Stable equilibria of the SimSiam-like dynamics in function space

$$U^*(x) = \sum_{i=1}^r f(s_i, \lambda) q_i u_i(x)$$

Diagram illustrating the components of the optimal predictor $U^*(x)$:

- Any $0 \leq r \leq k$** : Points to the upper limit r of the summation.
- Regularization**: Points to the parameter λ in the function $f(s_i, \lambda)$.
- Spectral filter**: Points to the function $f(s_i, \lambda)$.
- Canonical correlation $s_1 \geq s_2 \geq \dots$** : Points to the parameter s_i in the function $f(s_i, \lambda)$.

- Optimal predictor carries spectral information about an equilibrium:

maximized by non-collapsed equilibria

$$\text{maximized by non-collapsed equilibria} \rightarrow \text{Tr}(P^*) = \sum_{i=1}^r s_i \left(1 - \frac{\lambda}{f(s_i, \lambda)^2 + \lambda} \right)$$

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- Trace maximization of the optimal predictor as a representation learning mechanism ?

Predictive Embeddings through Inter-View Regressor Alignment

- Find optimal symmetrized regressor for each encoders pairs (U, V)

$$\frac{1}{2}\mathbb{E}[\|PU(x) - V(y)\|^2] + \frac{1}{2}\mathbb{E}[\|PV(y) - U(x)\|^2] + \frac{\lambda}{2}\|P\|^2$$

$$P_{U,V}^{\star} = \arg \min_{P \in \mathbb{R}^{k \times k}} \mathcal{L}_{reg}(P; U, V)$$

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- Learn encoders by minimizing regularized negative trace of optimal regressor:

Negative trace minimization to align representation

Regularization to fix the scale

$$\mathcal{L}_{PEIRA}(U, V) = -\frac{1}{2}\text{Tr}(P_{U,V}^{\star}) + \frac{\lambda}{2}\mathbb{E}[\|V(x)\|^2 + \|U(y)\|^2]$$

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What does it learn? Does it avoid collapse? Practical algorithms? Does it work?

PEIRA objective: Global maximizers

Theorem (Informal): Assume the data distribution is regular enough (existence of non-linear CCA).

Then: 1- Global minimizers of $\mathcal{L}_{PEIRA}$ are exactly:

$$(U^\star(x), V^\star(y)) = \sum_{i=1}^k f(s_i, \lambda) q_i (u_i(x), v_i(y))$$

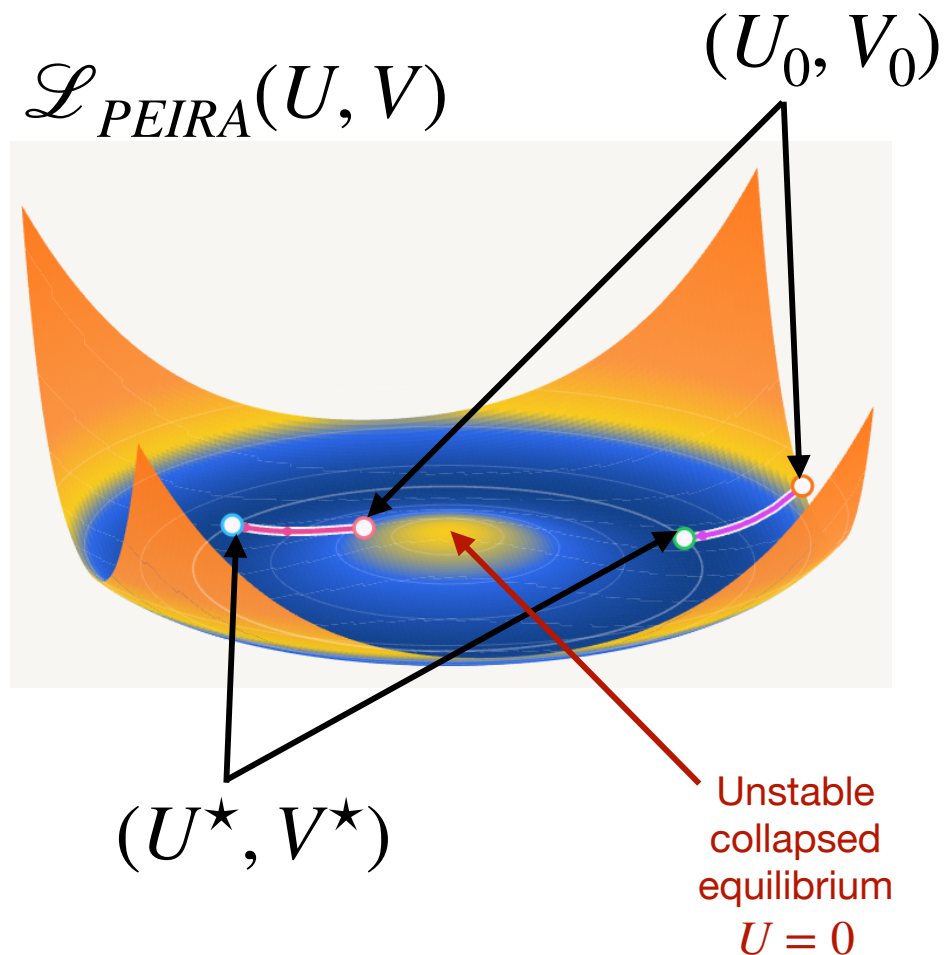
Encoder's dimension $\rightarrow k$

Any orthonormal family $(q_i)_{1 \leq i \leq r}$ of $\mathbb{R}^k$

Canonical functions $(u_i(x), v_i(y))$

Soft-thresholding filter $\frac{1}{\sqrt{2}} \left(\max(\sqrt{s} - \lambda), 0 \right)^{\frac{1}{2}}$

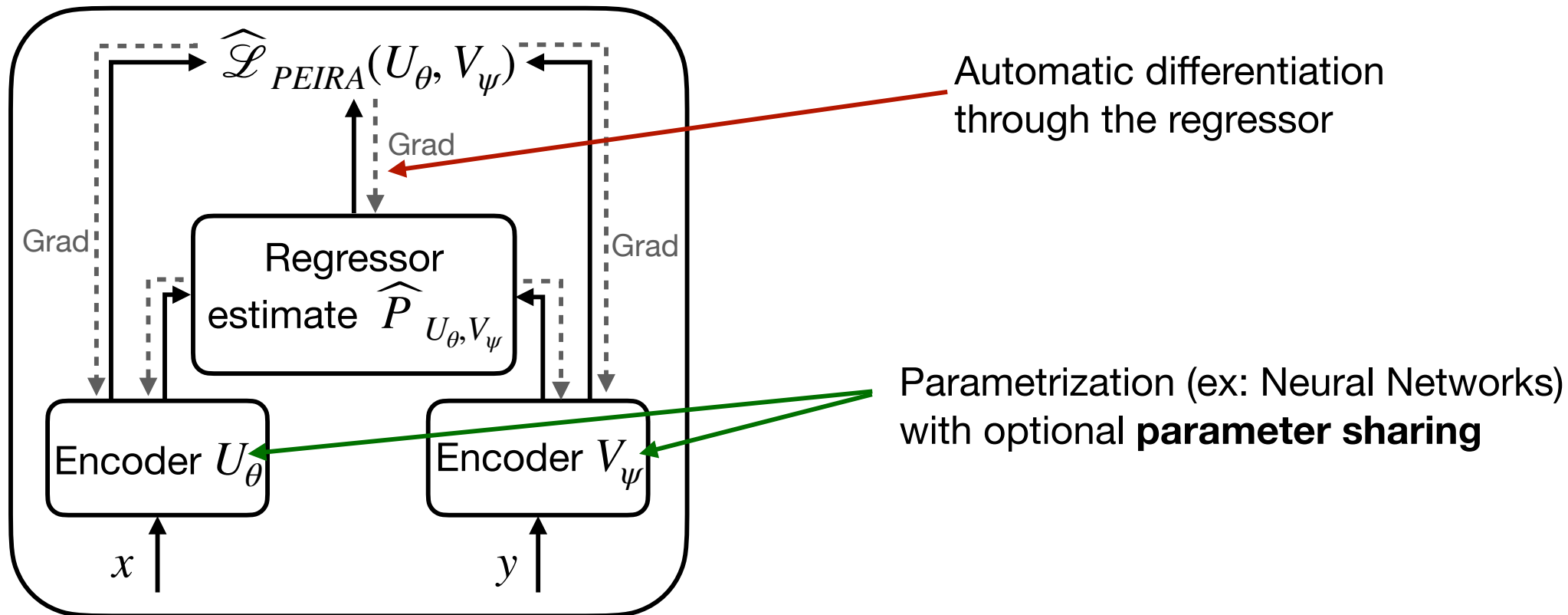
PEIRA objective: Stability and non-collapse



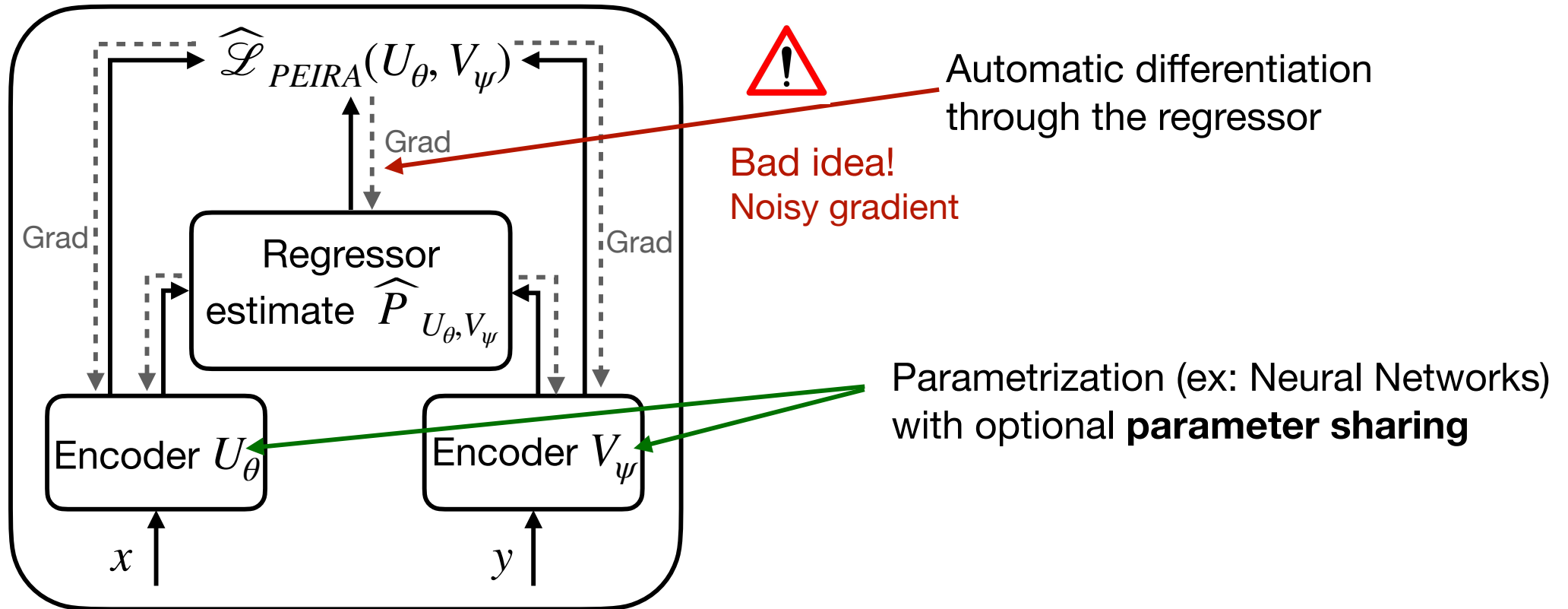
Theorem (Informal): Under the same assumptions as before:

- 1- Only global minimizers are stable.
- 2- All other critical points are unstable.

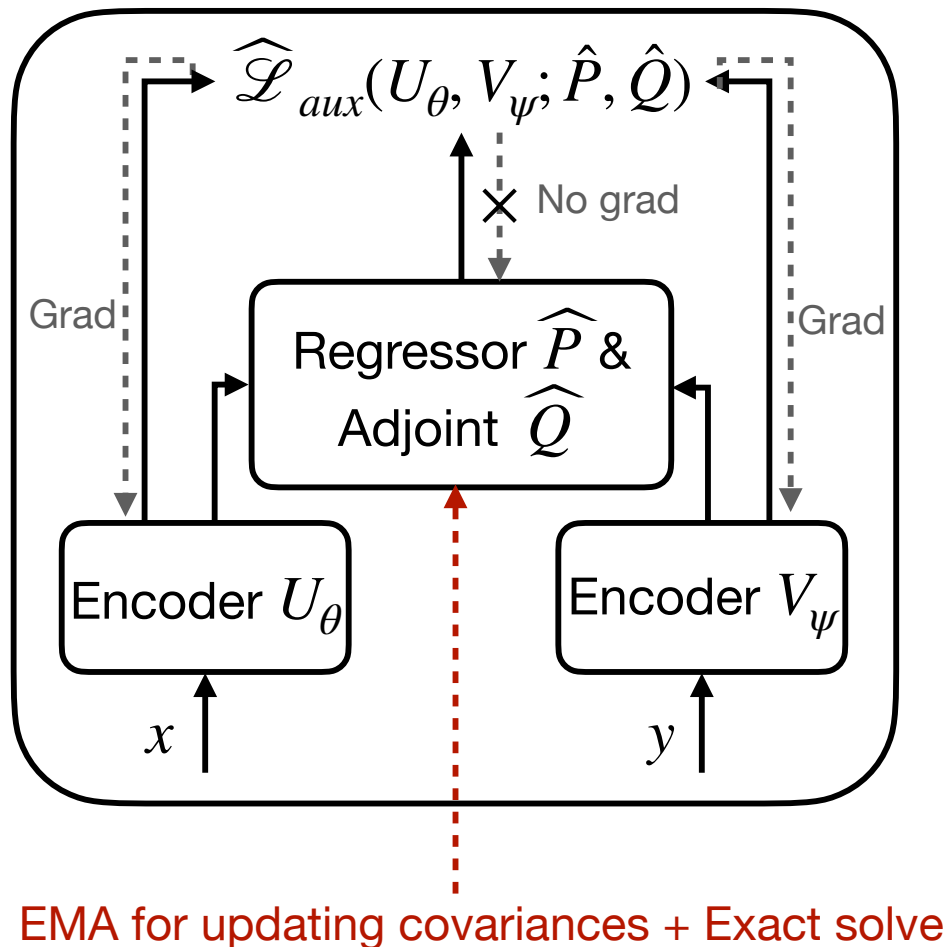
Optimizing PEIRA objective: A "bad" Algorithm



Optimizing PEIRA objective: A "bad" Algorithm



Optimizing PEIRA objective: A better Algorithm



Proposition:

$\mathcal{L}_{PEIRA}(U, V)$ has the same gradient as that of an auxiliary loss $\mathcal{L}_{aux}(U, V; P, Q)$ if:

- P is the optimal regressor
- Q satisfies

$$Q = (\mathbb{E} [U(x)U(x)^\top + V(y)V(y)^\top] + \lambda I)^{-1}$$

The auxiliary loss:

$$\begin{aligned} \mathcal{L}_{aux}(U, V; P, Q) = & \frac{1}{2} \mathbb{E} \left[QU(x)^\top (PU(x) - V(y)) \right] \\ & + \frac{1}{2} \mathbb{E} \left[QV(y)^\top (PV(y) - U(x)) \right] \\ & + \frac{\lambda}{2} \mathbb{E} [\|U(x)\|^2 + \|V(y)\|^2] \end{aligned}$$

How does PEIRA do in practice?

Tasks

SSL for natural images

Pre-training:

- Cifar10
- Imagenet

Downstream task:

- Classification Cifar10
- Classification Imagenet1k

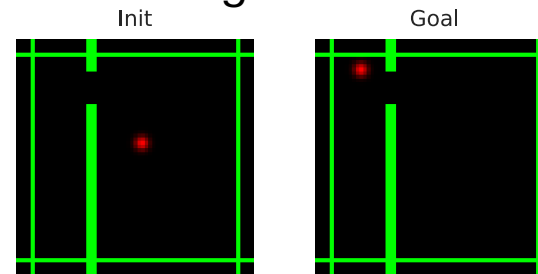
SSL for world modeling

Pre-training: Action conditioned video

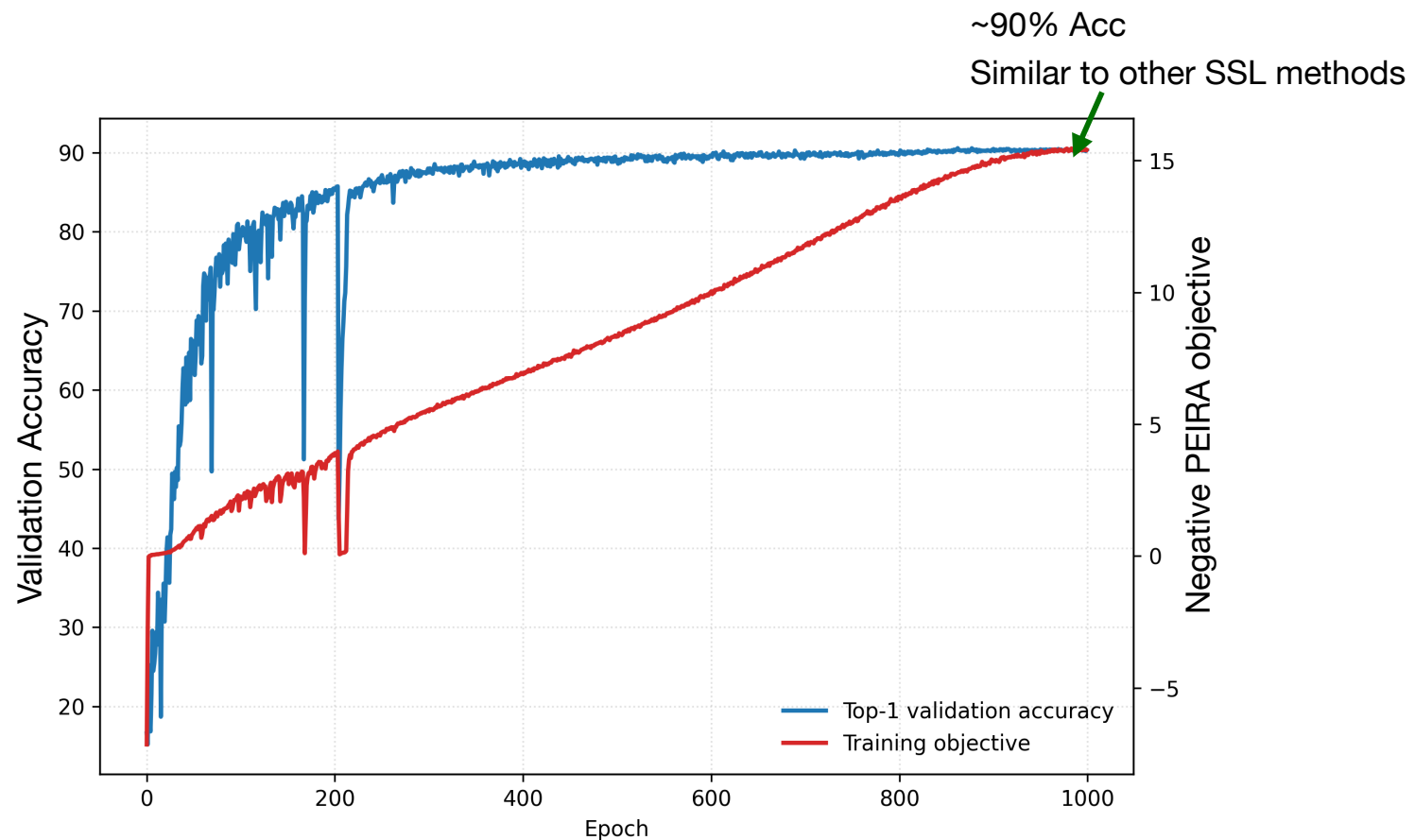
- Two rooms environment

Downstream task:

- Planning from initial to goal state

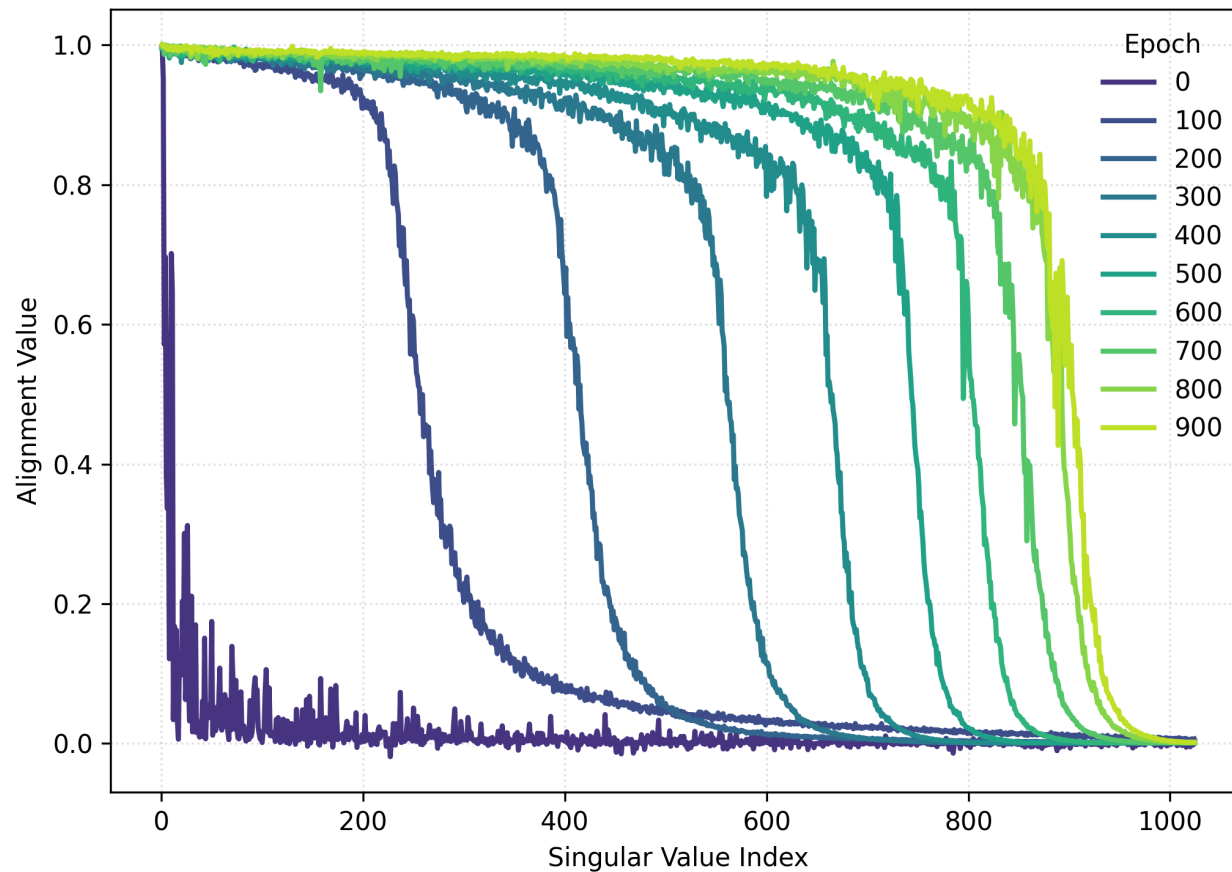


SSL for images: Cifar10



Strong correlation between Val acc and PEIRA objective

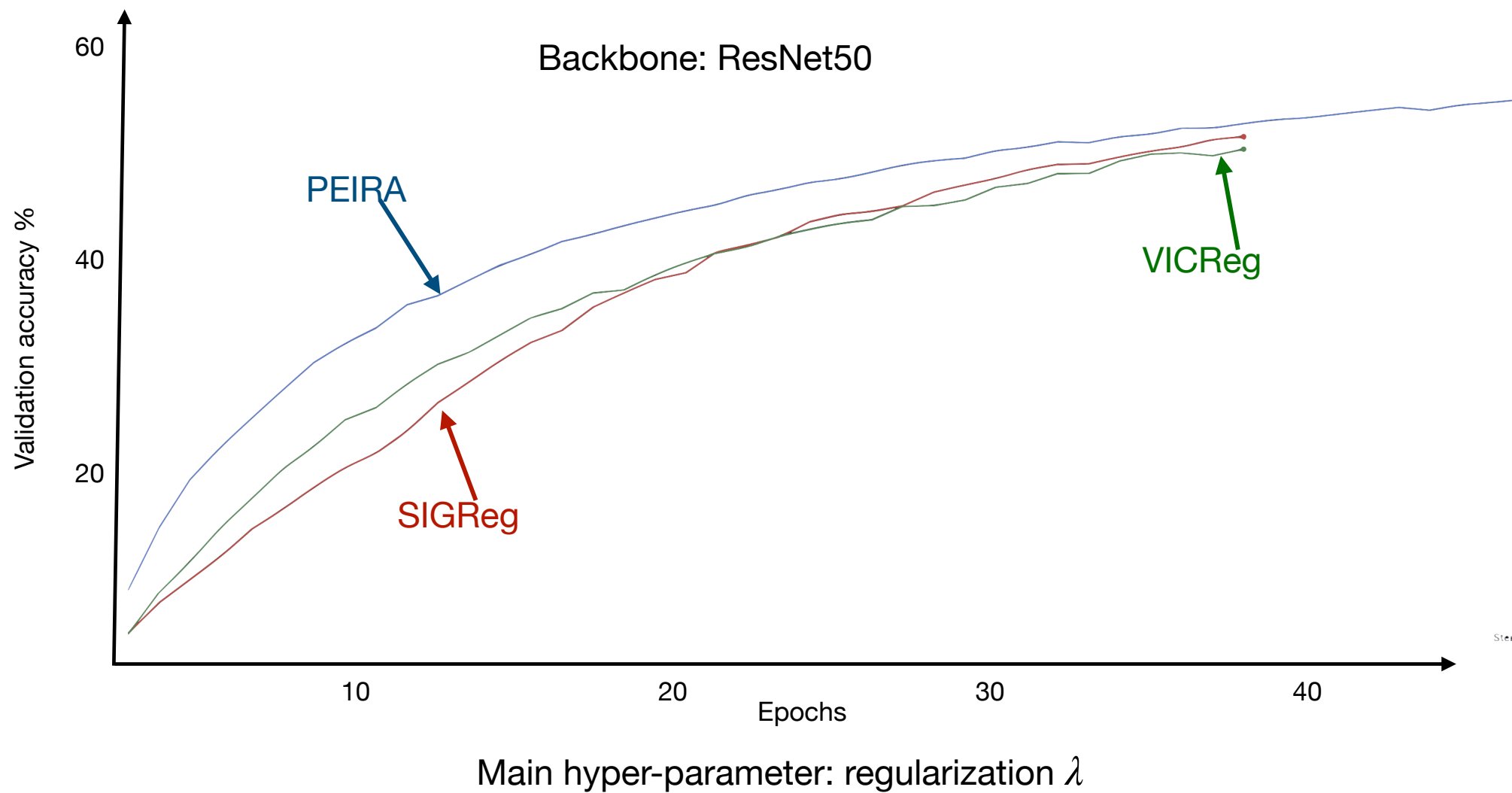
SSL for images: Cifar10



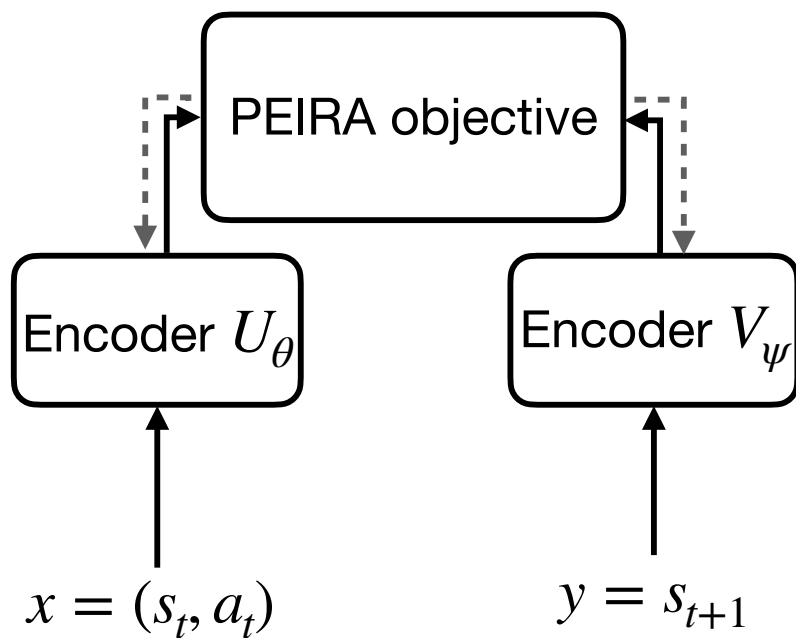
Gradual alignment between regressor and cross-covariance

Similar to Tian et al. 2022

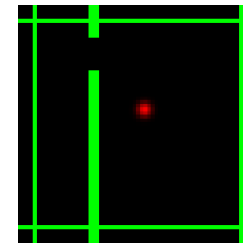
SSL for images: Preliminary results on ImageNet1k



Self-supervised World models: Two rooms



- State s_t : Image



- Action a_t : Up, down, left, right

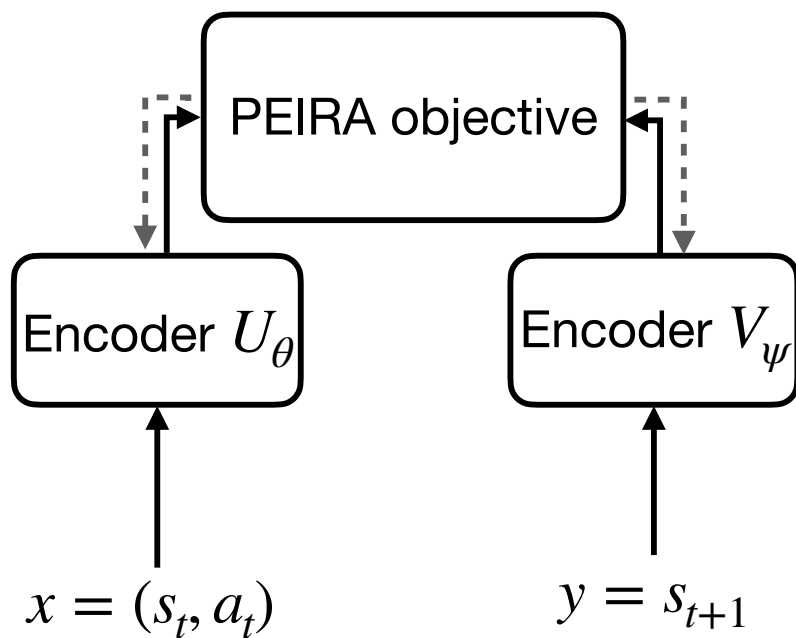
- State encoder V_ψ : Impala

- State action encoder $U_\theta(s, a) = RNN_\phi(V_\psi(s), a)$

$$\theta = (\phi, \psi)$$

Parameter sharing

Self-supervised World models: Two rooms



Setup: Joint pre-training of predictor and encoder

Comparison:

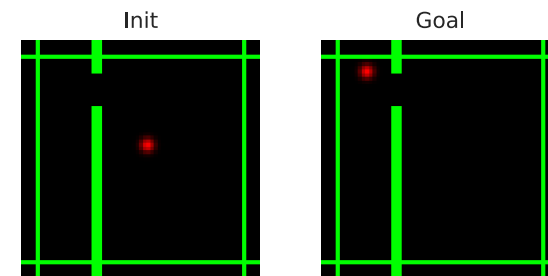
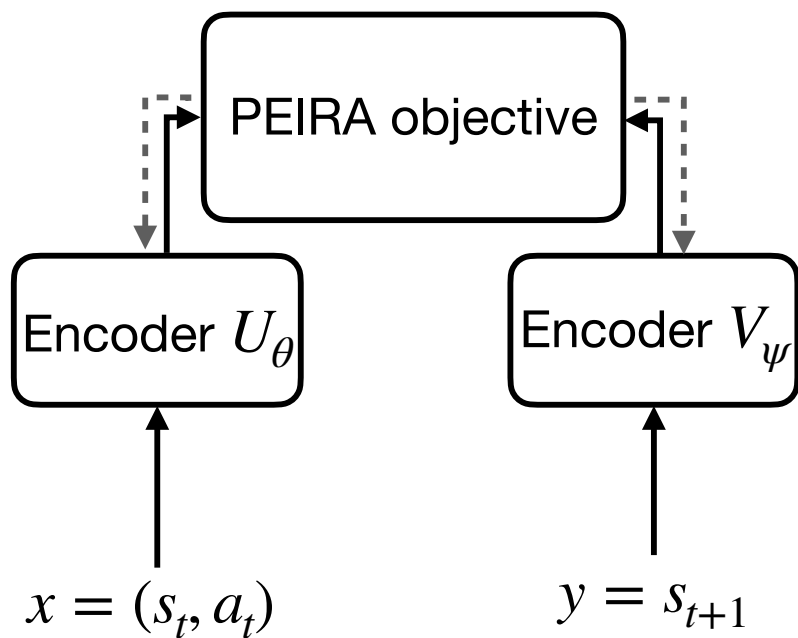
- The objective from PLDM (Sobal et al. 2025)

$$\mathcal{L}_{PLDM} = \mathcal{L}_{pred} + \alpha \mathcal{L}_{var} + \beta \mathcal{L}_{cov} + \delta \mathcal{L}_{sim} + \omega \mathcal{L}_{IDM}$$

- Le WorldModel (Maes et al. 2026)

$$\mathcal{L}_{LeWM} = \mathcal{L}_{pred} + \alpha \mathcal{L}_{Sigreg}$$

Self-supervised World models: Two rooms



Method	Success Rate (%)
$\mathcal{L}_{PEIRA}$	80%
$\mathcal{L}_{LeWM}$	87%
$\mathcal{L}_{PLDM}$	97%
$\mathcal{L}_{PLDM} \text{ w/o } \mathcal{L}_{var}$	47%
$\mathcal{L}_{PLDM} \text{ w/o } \mathcal{L}_{cov}$	46%
$\mathcal{L}_{PLDM} \text{ w/o } \mathcal{L}_{sim}$	61%
$\mathcal{L}_{PLDM} \text{ w/o } \mathcal{L}_{IDM}$	1%

Preliminary but encouraging

Conclusion

Summary

- There is a formal connexion between SimSiam-like SSL methods and non-linear CCA
- Collapse is still an issue for these methods
- PEIRA leverages this connection with CCA while provably avoiding collapse
- Promising results on image representation and world modeling

Future directions/open questions

- Similar connexion for Dino? SIGReg? etc
- More applications? Video, time series
- Smarter algorithms? Avoiding full matrix inversion? Variance reduction, etc

Thank you!